

Design and Implementation of Inverse Kinematics Algorithm to Manipulate 5-DOF Humanoid Robotic Arm

Ammar A. Al-Hamadani
Dep. of Network Engineering
Al-Iraqia University
Baghdad, Iraq.
ammr.aladin@aliraqia.edu.iq

Mohammed Z. Al-Faiz
Alhussain University College
Baghdad, Iraq.
mzalfaiz12@huciraq.edu.iq

Abstract—Inverse Kinematics (IK) is a mathematical approach to compute joint angles of an arm based on given position and orientation of the wrist in the space. This paper presents implementation of new analytical solution of Inverse Kinematics algorithm to manipulate new 3D printed 5-DOF Humanoid Robotic Arm termed (HRA). The proposed IK algorithm is programmed using MATLAB to find the joints angles which used to control the servos of HRA via Arduino microcontroller. Six desired positions are used as targets to be achieved on experimental workspace in the LAB. The IK algorithm is assessed by calculating the Root Mean Squared Error (RMSE) for the absolute error vector for six positions. Results shows that minimum achieved (RMSE) for the arm's end effector position is 0.5774.

Keywords—Humanoid Robotic Arm; Inverse Kinematics; Analytical Solution; Brain Computer Interface.

I. INTRODUCTION

Brain Computer Interface (BCI) is the state of the art system that harnesses the brain waves of the human to control outside world machine such as robotic arms [1][2][3]. One of the main concepts in the field of robotic manipulator system is the kinematic chain of the arm joints [4]. The kinematics analysis includes the forward (direct) and inverse kinematics. The Forward Kinematics (FK) is used to compute the position and orientation of the wrist End-Effector (EE) based on given joint angles (joint variables) of the arm. The Inverse Kinematics (IK) is used to compute the joint angles of the arm based on given position and orientation of the wrist (EE) in the space. Many approaches provided solution for the IK such as: numerical, iterative, algebraic, and geometrical solutions [5]. The model of robotic arm joints should take considerable attention since it's directly affect the success of arm movement. One of the important concepts used in the modeling of robotic arms joints is the Denavit-Hartenberg (D-H) method. The main concept of such method is to model the arm as series of joint and assign a reference joint to start the motion. The motion is transformed from reference to last joint and all the transformations are formulated in total transformation matrix that represent the FK of the arm [6]. There are many researches in the literature that adopted the FK/IK algorithms to control robotic manipulators. The paradigm presented in [7] used D-H technique to model robotic arm for rehabilitation of motor dysfunction patients, AAN controller has been used based on IK algorithm. The researchers in [8] used three approaches to model 5-DOF robotic arm which are D-H, screw theory, and iterative methods. Each of those approaches was used in the analytical solution of IK and results showed that D-H method achieved the fastest solution for the analytical IK which considered good advantage in the real-time applications. In [9], analytical solution based on D-H technique was proposed to

manipulate 5-DOF Humanoid Robotic Arm. The researchers used Graphical User Interface (GUI) under MATLAB to simulate the motion of arm using the proposed IK. From these previous researches, it is found that analytical solution method based on D-H parameters gives good and fast approximations of the required joint angles. However, no previous research dealt with solutions that fits to the design requirement of our case. The problem definition of this research is to implement a human like robotic arm with kinematic model that is capable to be controlled in on-line mode by the BCI system. In this paper, implementation of new analytical solution of IK was used to manipulate new 3D printed, 5-DOF Humanoid Robotic Arm (HRA). Arduino mega microcontroller was used to interconnect a designed MATLAB program of the IK algorithm with the HRA's servo motors to deliver the computed joints angles. The accuracy of the paradigm was tested in terms of positional error of (EE) by computing the Root Mean Squared Error (RMSE) for the absolute error vector of the reached position.

II. PROPOSED METHOD

A. Forward and invers kinematics

The kinematic chain of the HRA was defined based on Denavit-Hartenberg (D-H) technique where the direction of joints rotation is around z-axis and termed as ($\theta_1, \theta_2, \theta_3, \theta_4, \theta_5$). The displacement between two joints called the common normal and represents the x-axis and termed as (a_3, d_5). The kinematic scheme of the constructed 5-DOF HRA with its system of axis is depicted in Fig. 1

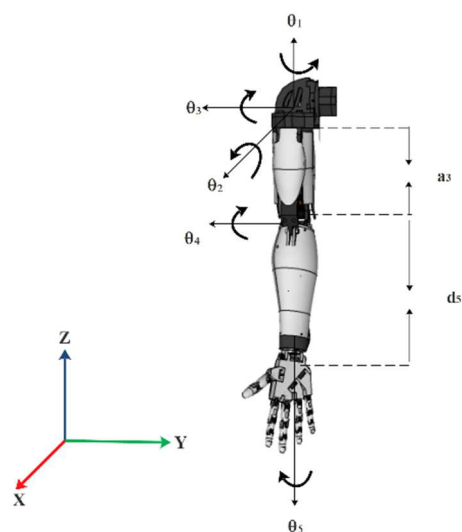


Figure 1: Kinematic Model of 5-DOF Humanoid Robotic Arm.

The D-H parameter corresponding to the links configuration of the designed manipulator are presented in table I, such parameters are related to our designed robotic

arm only. where, θ_i represents rotation about Z_i , d_i is the z-axis displacement, a_i is the displacement between Z_{i-1} and Z_i , α_i is the rotation about x-axis and hence that, $a_3 = -35\text{cm}$ and $d_5 = 29\text{cm}$. These parameters are utilized to describe the position and orientation of joint-link using a number of transformation matrices equal to the number of joints given in Eqs. (2-6). These matrices representing the Forward Kinematics (FK) matrices that transfer the movement from joint to another and they can be denoted as (A_i) in Eq. (1), where, $(i = 1, 2, \dots, 5)$. Keeping in mind that $\cos\left(\theta + \frac{\pi}{2}\right) = -\sin(\theta)$ and, $\sin\left(\theta + \frac{\pi}{2}\right) = \cos(\theta)$.

TABLE I. D-H PARAMETERS TABLE FOR DESIGNED HRA

Frame No. (i)	θ_i	d_i	a_i	α_i
1	$\theta_1 + \frac{\pi}{2}$	0	0	$\frac{\pi}{2}$
2	$\theta_2 + \frac{\pi}{2}$	0	0	$-\frac{\pi}{2}$
3	θ_3	0	a_3	0
4	$\theta_4 + \frac{\pi}{2}$	0	0	$-\frac{\pi}{2}$
5	θ_5	d_5	0	0

$$A_i = \begin{bmatrix} C\theta_i & -S\theta_i C\alpha_i & S\theta_i S\alpha_i & a_i C\theta_i \\ S\theta_i & C\theta_i C\alpha_i & -C\theta_i S\alpha_i & a_i S\theta_i \\ 0 & S\alpha_i & C\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

$$A_1 = \begin{bmatrix} -S_1 & 0 & C_1 & 0 \\ C_1 & 0 & S_1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$$A_2 = \begin{bmatrix} -S_2 & 0 & -C_2 & 0 \\ C_2 & 0 & -S_2 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

$$A_3 = \begin{bmatrix} C_3 & -S_3 & 0 & a_3 C_3 \\ S_3 & C_3 & 0 & a_3 S_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$$A_4 = \begin{bmatrix} -S_4 & 0 & -C_4 & 0 \\ C_4 & 0 & -S_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (5)$$

$$A_5 = \begin{bmatrix} C_5 & -S_5 & 0 & 0 \\ S_5 & C_5 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

where, C is referred to $\cos(\theta)$ and S is referred to $\sin(\theta)$.

The total FK transformation matrix from joint one to joint five (T_5^1) can be calculated as in Eq.7:

$$T_5^1 = A_1 A_2 A_3 A_4 A_5 = \begin{bmatrix} n_x & o_x & a_x & P_x \\ n_y & o_y & a_y & P_y \\ n_z & o_z & a_z & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (7)$$

where, $n_x, n_y, n_z, o_x, o_y, o_z, a_x, a_y, a_z$ are rotation parameters of (EE) around (x,y,z) axis, P_x, P_y, P_z are the position parameters thus, T_5^1 as whole matrix represents the (EE) configuration of the HRA. For the 5-DOF arm, Inverse Kinematic (IK) problem is required to be solved to get the five joints angles ($\theta_1, \theta_2, \dots, \theta_5$) to locate the desired position and orientation of the EE. Now, to define the IK problem, suppose that the desired position of the HRA is defined by Eq. 8. Orientation (rotation) of HRA can be defined by knowing the desired rotation of (EE) about x, y, z axis respectively. Such rotations was represented in Eq.9.

$$P^d = [P_x^d \ P_y^d \ P_z^d]^T \quad (8)$$

$$R^d = \begin{bmatrix} C_\rho C_\psi & C_\rho S_\psi S_\phi - S_\rho C_\phi & C_\rho S_\psi C_\phi + S_\rho S_\phi \\ C_\rho C_\psi & S_\rho S_\psi S_\phi + C_\rho C_\phi & S_\rho S_\psi C_\phi - C_\rho S_\phi \\ -S_\psi & C_\psi S_\phi & C_\psi C_\phi \end{bmatrix} = \begin{bmatrix} r_{11}^d & r_{12}^d & r_{13}^d \\ r_{21}^d & r_{22}^d & r_{23}^d \\ r_{31}^d & r_{32}^d & r_{33}^d \end{bmatrix} \quad (9)$$

where, P^d is the desired position vector and P_x, P_y, P_z are the position parameters, R^d is the desired rotation (orientation) matrix, ρ, ψ, ϕ are the desired rotation angles of EE about x, y, z axis respectively with r_{ij}^d are desired rotation parameter. The desired homogenous transformation matrix T^d can then be formed as follows:

$$T^d = \begin{bmatrix} R^d & P^d \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11}^d & r_{12}^d & r_{13}^d & P_x^d \\ r_{21}^d & r_{22}^d & r_{23}^d & P_y^d \\ r_{31}^d & r_{32}^d & r_{33}^d & P_z^d \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} n_x^d & o_x^d & a_x^d & P_x^d \\ n_y^d & o_y^d & a_y^d & P_y^d \\ n_z^d & o_z^d & a_z^d & P_z^d \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (10)$$

It's assumed that the value of the wrist angle (θ_5) is always zero because it represents the end effector of the HRA. That leads to the fact that its value doesn't affect the pose of the HRA. For all calculations, the following joint constraints (specifically related to our designed HRA) shown in table II were followed and any joint angle out of these constraints are designed to be excluded by the program.

TABLE II. HRA JOINT ANGLES CONSTRAINTS.

Joint	Range of Usable Angles
1	$-90 \leq Q1 \leq 90$
2	$-90 \leq Q2 \leq 0$
3	$0 \leq Q3 \leq 180$
4	$0 \leq Q4 \leq 90$
5	$Q5 = \text{zero}$

Using the matrix elements provided in Eq. 10 and the D-H parameters in table I, the solution of the joint angles is given directly as follows:

$$\theta_1 = \tan^{-1}\left(\frac{-o_x}{o_y}\right) \quad (11)$$

$$\theta_2 = \tan^{-1}\left(\frac{P_x S_1 - P_y C_1}{P_z}\right) \quad (12)$$

$$\theta_3 = \tan^{-1}\left(\frac{S_3}{C_3}\right), \text{ where, } C_3 = \frac{P_z - a_z d_5}{a_3 C_2} \text{ and } S_3 = \sqrt{1 - C_3^2} \quad (13)$$

$$\theta_4 = \pi - \cos^{-1}\left(\frac{a_3^2 + d_5^2 - SE^2}{2 a_3 d_5}\right), \text{ where, } SE = \sqrt{P_x^2 + P_y^2 + P_z^2} \quad (14)$$

The positional error of (EE) can be checked using these angles then calculate the Root Means Square Errors (RMSE) of the absolute error vector between the resultant actual position and the desired one as follows:

$$\begin{bmatrix} err_x \\ err_y \\ err_z \end{bmatrix} = \begin{bmatrix} P_x - P_x^a \\ P_y - P_y^a \\ P_z - P_z^a \end{bmatrix} \quad (15)$$

$$RMSE = \sqrt{\frac{err_x^2 + err_y^2 + err_z^2}{3}} \quad (16)$$

where, $[err_x, err_y, err_z]^T$ is the absolute error vector in x, y, z axis respectively, $[P_x, P_y, P_z]^T$ is the desired position vector, and $[P_x^a, P_y^a, P_z^a]$ is the actual position vector.

B. Humanoid Robotic Arm (HRA)

The proposed HRA is 5-DoF right arm which built based on InMoov design using 3D printer. The arm consisted of six parts, palm with five fingers, rotating wrist, forearm, bicep, shoulder, and torso. The five fingers and wrist are actuated by six servo motors of type (Tower Pro MG995), the forearm, bicep, and shoulder are actuated using four servo motor of type (Hitec HS-805BB). The Arduino microcontroller controls servo motors via C++ program. The total maximum current consumption can reach up to 36 A at any moment depending on the room temperature and the weight of the object that the HRA is carrying. Therefore, External power supply of 40 A was used and fixed to the back of HRA. 3D printed Arduino support was used as frame to attach the Arduino to the back of HRA. Snap shoot of the designed HRA is presented in Fig. 2.

C. Workspace and layout

To form a working space area that can be used to perform and measure the IK results, it was proposed to design a layout of $(2 \times 2) m$ in order to place the HRA on it. The layout consisted of four quadrants coordinate system where each quadrant is of $(1 \times 1) m$ as $(x \times y)$ axis. The point where the arm attached to the torso of the proposed HRA was regarded as the starting point of movement and thus, placed at the origin of the coordinates of layout as in Fig.3. The HRA was mounted on stand of height 72 cm which made the arm's origin rises by 72 cm of the ground thus, the z -axis of the arm works in negative zone $(-z)$. The selected working area of the HRA was proposed to be in the fourth quadrant of the layout so that, x -axis is positive, y -axis is negative, and keeping in mind that z -axis is negative too. Noting that, the exact working area is between 64 cm in x -axis and -64 cm in y -axis.

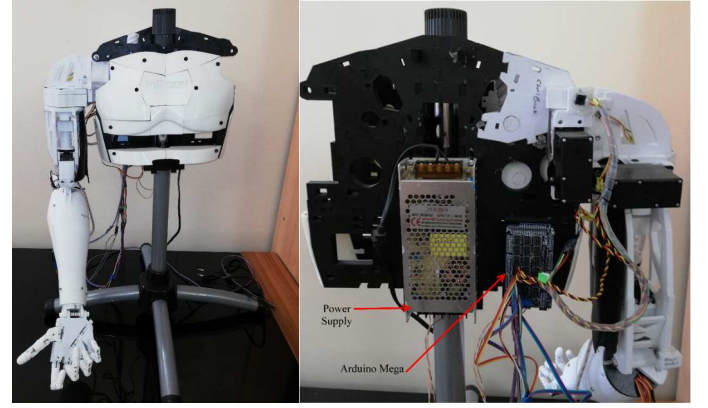


Figure 1: Designed Humanoid Robotic Arm (HRA).

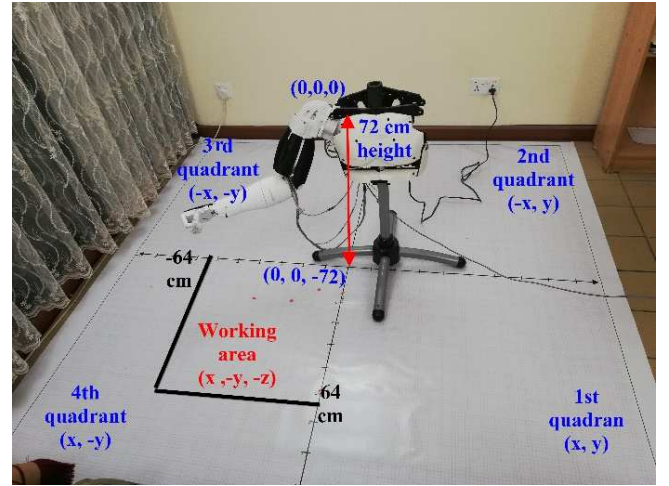


Figure 2: Layout Define the Place and the Working Area of HRA.

III. RESULTS AND DISCUSSION

To calculate the required joint angles from desired position and rotation of the EE, the equations (11-14) were used. Set of desired positions along with their desired rotations were suggested to calculate the required joint angles for each case. The accuracy of the IK algorithm was computed using Eqs.15 and 16 where, the actual position was calculated by using the joint angles provided from the IK solution as input to the FK algorithm with the aid of D-H parameters in table I. The actual position calculated from the FK algorithm was compared against the desired one and then the IK accuracy can be investigated based on the resultant RMSE of the absolute position error vector, as shown in table III. Table III shows the position number along with its desired position, desired rotation angles (ρ, ψ, ϕ) , and desired rotation matrix given by Eq.9, these items represents input parameters to IK algorithm. The resultant θ represents by (calculated angles using IK and are computed using Eqs. 15-16. Also the table shows the actually reached position on the layout (measured reached position) achieved after feeding the calculated θ to our FK algorithm. The position error is firstly calculated as position error vector using Eq.15 then, the RMSE is computed to estimate the overall reaching accuracy of IK algorithm. As can be seen from table III, there is a rise of RMSE error especially in the first and fifth positions. The second factor is associated with analytical solution of IK that effected by the non-linearity of trigonometric functions.

TABLE III. RESULTS OF COMPUTING THE REQUIRED θ (JOINT ANGLES) AND POSITION ERROR FOR A SET OF DESIRED POSITIONS AND ORIENTATIONS.

No.	Desired Position (cm)	Desired rotation angles (ρ, ψ, ϕ) (degree)	Desired rotation matrix Eq. 9	Calculated angles (using IK) (degree) Eqs. (15-16)	Measured reached position (cm)	Position error vector (cm) Eq. 15	RMSE Eq. 16
1	$\begin{bmatrix} 64 \\ 0 \\ 0 \end{bmatrix}$	(0, 90, 0)	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ 90 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 60 \\ 0 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$	2.3094
2	$\begin{bmatrix} 15 \\ -10 \\ -59 \end{bmatrix}$	(-147, 30, -168)	$\begin{bmatrix} -0.8572 & 0 & 0.5150 \\ -0.0894 & 0.9848 & -0.1488 \\ -0.5072 & -0.1736 & -0.8441 \end{bmatrix}$	$\begin{bmatrix} 0 \\ -9.62 \\ 0 \\ 31 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 15 \\ -10 \\ -60 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	0.5774
3	$\begin{bmatrix} 20 \\ -20 \\ -50 \end{bmatrix}$	(-168, 50, -152)	$\begin{bmatrix} -0.6355 & 0.1651 & 0.7542 \\ -0.1385 & 0.9366 & -0.3218 \\ -0.7595 & -0.3090 & -0.5724 \end{bmatrix}$	$\begin{bmatrix} -10 \\ -17.97 \\ 0 \\ 52.56 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 21 \\ -20 \\ -52 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$	1.291
4	$\begin{bmatrix} 15 \\ 0 \\ -60 \end{bmatrix}$	(180, 31, 180)	$\begin{bmatrix} -0.8572 & 0 & 0.5150 \\ 0 & 1 & 0 \\ -0.5150 & -0.8572 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 29.94 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 15 \\ 0 \\ -61 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$	0.5774
5	$\begin{bmatrix} 21 \\ -35 \\ -48 \end{bmatrix}$	(-161, 24, -140)	$\begin{bmatrix} -0.866 & 0 & 0.5 \\ -0.2939 & 0.809 & -0.509 \\ -0.4045 & -0.5878 & -0.7006 \end{bmatrix}$	$\begin{bmatrix} 0 \\ -36.098 \\ 11.81 \\ 20.29 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 21 \\ -35 \\ -52 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$	2.3094
6	$\begin{bmatrix} 50 \\ -20 \\ -30 \end{bmatrix}$	(-138, 56, -124)	$\begin{bmatrix} -0.3123 & 0.1632 & 0.9359 \\ -0.2804 & 0.9254 & -0.2549 \\ -0.9077 & -0.342 & -0.2432 \end{bmatrix}$	$\begin{bmatrix} -10 \\ -20.16 \\ 45.7 \\ 31.33 \\ 0 \end{bmatrix}$	$\begin{bmatrix} 50 \\ -18 \\ -32 \end{bmatrix}$	$\begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$	1.633

As an example of how to measure the reached position of the HRA on the workspace, position no. 2 taken from table III is previewed as snapshots in Fig. 4. The red point on the workspace layout represent the projected position that the HRA reached after executing the obtained θ . The height of the point was measured using tape measure where the measured level was taken up to the middle of the palm.

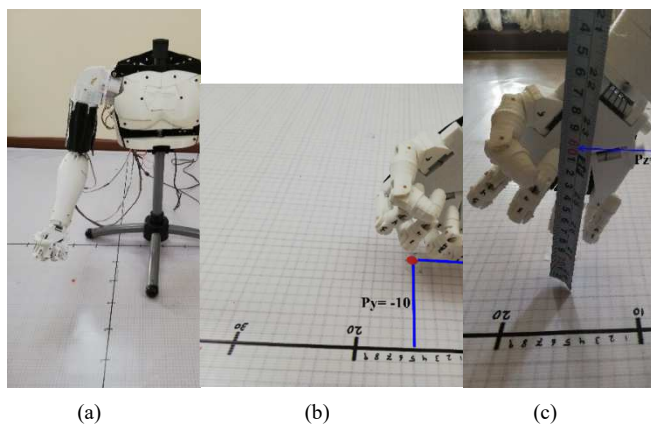


Figure 3: Measurements of the Reached Position (15, -10, -60)
(a) Front View, (b) Measurement of x and y axis, (c) Measurement of z-axis.

IV. CONCLUSION

This work is a part of BCI research to manipulate Humanoid Robotic Arm (HRA) using Inverse Kinematics (IK) algorithm. New and specifically derived IK algorithm was successfully implemented using 3D printed humanoid robotic arm. The inverse kinematics problem of the arm can be stated as follows: given the position and the orientation of the hand, find the joint angles that lead the arm to reach these position and orientation. The experiment includes choosing set of points to compute their corresponding joint angles (i.e. θ). These joint angles are calculated using derived equations (11-14) based on desired position (Eq.8) and orientation of EE (Eq.9). These θ s are then passed to the HRA from MATLAB via USB and Arduino microcontroller that hosting the control of the servos. The aim of this experiment was to assess the accuracy of the HRA for reaching the desired point as a proactive step to be integrated with Brain Computer Interface (BCI) system in another research. Lowest practical (RMSE) positional error was 0.5774 achieved in the positions (15, -10, -59) and (15, 0, -60) respectively. Highest practical (RMSE) was 2.3094 happened in the positions (64, 0, 0), and (21, -35, -48) respectively. The higher RMSE was likely caused by the dynamic issues associated with the servomotors in addition to the non-linearity of trigonometric functions. So, a degree of error ratio between the calculated θ and the actual rotation of the servos was appeared. Therefore; as future work, it's suggested to use an optimization technique with the proposed inverse kinematics

algorithm to reduce this error. The designed algorithm is suggested to be used in the application of BCI to help amputated persons.

To compare our proposed technique with other recent techniques, the strategy presented in this article depended on specifically derived IK algorithm (MATLAB code) that communicate with HRA via Arduino MEGA microcontroller in a form of character code 'char.'. While, the work presented in [7] uses AAN strategy of training to control the robotic arm. On the other hand, researchers in [10] used ASED algorithm based on PID controller to communicate with the robotic arm. In this case, our work regarded as open source robotic arm that can be cooperated with any controller as well as bio signals such as EEG or EMG.

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